Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
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Recitation 2 Solutions

Problem 2.1

Suppose every child is equally likely to be a boy or a girl, and that the gender of any newborn is independent of genders of all previously born children. A family has two children.[1]

(a) Determine the probability that both children are boys.

Solution:

Let $\mathcal{A}_1$ be the event that the first child is a boy and $\mathcal{A}_2$ be the event that the second child is a boy. $\mathcal{A}_1$ and $\mathcal{A}_2$ are independent and $\mathbb{P}(\mathcal{A}_1) = \mathbb{P}(\mathcal{A}_2) = 1/2$. Hence,

$$\mathbb{P}(\mathcal{A}_1 \cap \mathcal{A}_2) = \mathbb{P}(\mathcal{A}_1)\,\mathbb{P}(\mathcal{A}_2) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

(b) Determine the probability that both children are boys given that at least one child is a boy.

Solution:

We can represent the sample space as a 2×2 table:

First \ Second	**Boy**	**Girl**
Boy	**1/4**	1/4
Girl	1/4	1/4

We define $\mathcal{A}$ to be the event that both children are boys and $\mathcal{B}$ to be the event that at least one child is a boy. Event $\mathcal{A}$ includes just the top left outcome in the table (bold) and event $\mathcal{B}$ includes the top row and the left column (gray). We obtain

$$\mathbb{P}(\mathcal{A} \mid \mathcal{B}) = \frac{\mathbb{P}(\mathcal{A} \cap \mathcal{B})}{\mathbb{P}(\mathcal{B})} = \frac{\mathbb{P}(\mathcal{A})}{\mathbb{P}(\mathcal{B})} = \frac{1/4}{3/4} = \frac{1}{3}.$$

(c) Determine the probability that both children are boys given that least one child is a boy born on a Tuesday. Assume each child is equally likely to be born on any day of the week, independently of all other children.

[1] This problem was posed by Gary Foshee at a gathering of puzzle enthusiasts. You can find lots of discussion about it online—see this New Scientist article or this blog post.

Solution:

In this case, there are 14 equally likely outcomes for each child (two sexes and seven days of the week are assigned independently). One outcome corresponds to a boy born on Tuesday, six outcomes correspond to a boy born on another day of the week, and seven outcomes correspond to a girl.

We can represent the sample space as a 3×3 table:

First \ Second	**Boy on T**	**Boy on another day**	**Girl**
Boy on T	**1/196**	**6/196**	7/196
Boy on another day	**6/196**	**36/196**	42/196
Girl	7/196	42/196	49/196

As in part (b), $\mathcal{A}$ is the event that both children are boys. Event $\mathcal{A}$ includes the top left 2×2 sub-table (bold). As before, $\mathbb{P}(\mathcal{A}) = 1/4$, which can be easily verified from the table. We define $\mathcal{B}'$ to be the event that at least one child is a boy born on Tuesday. Event $\mathcal{B}'$ includes the top row and the left column of the table above (gray),

$$\mathbb{P}(\mathcal{B}') = \frac{1+6+6+7+7}{196} = \frac{27}{196} \approx 0.14,$$

which is substantially lower than $\mathbb{P}(\mathcal{B}) = 3/4$.

We now obtain

$$\mathbb{P}(\mathcal{A} \,|\, \mathcal{B}') = \frac{\mathbb{P}(\mathcal{A} \cap \mathcal{B}')}{\mathbb{P}(\mathcal{B}')} = \frac{1+6+6}{1+6+6+7+7} = \frac{13}{27} \approx 0.48.$$

Remark: This problem illustrates how additional information can affect our beliefs. In part (b), eliminating the outcome that corresponds to two girls increases the probability that both children are boys. This is intuitive: knowing that at least one child is a boy increases the probability that both children are boys. Part (c) seems counterintuitive at first since sex and birthday are independent. But this "intuitive" thinking leads to a wrong conclusion. A more careful analysis reveals that changing the conditioning event leads to a more fine-grained sample space and changes the resulting conditional probability.

Problem 2.2

Your problem set is due in 15 minutes! It is in one of n drawers, but they are messy, and you're not sure which drawer it is in. The probability that the problem set is in drawer k is p_k for $k = 1, \ldots, n$. If the problem set is in drawer k and you search there, you will find it with probability q_k for $k = 1, \ldots, n$. Suppose you searched in drawer i and did not find the problem set.

(a) Determine the probability that the problem set is in drawer j for any $j \neq i$.

Solution:

Let $\mathcal{A}_k$ be the event that the problem set is in drawer $k = 1, \ldots, n$, so $\mathbb{P}(\mathcal{A}_k) = p_k$. Events $\mathcal{A}_1, \ldots, \mathcal{A}_n$ form a partition of the sample space. Let $\mathcal{B}_k$ be the event that you find the problem set in drawer $k = 1, \ldots, n$, so $\mathbb{P}(\mathcal{B}_k \,|\, \mathcal{A}_k) = q_k$. We use Bayes' rule to determine

$$\mathbb{P}(\mathcal{A}_j \,|\, \mathcal{B}_i^{\mathrm{c}}) = \frac{\mathbb{P}(\mathcal{A}_j)\ \mathbb{P}(\mathcal{B}_i^{\mathrm{c}} \,|\, \mathcal{A}_j)}{\mathbb{P}(\mathcal{B}_i^{\mathrm{c}})}.$$

For $j \neq i$, $\mathbb{P}(\mathcal{B}_i^{\mathrm{c}} \,|\, \mathcal{A}_j) = 1$ as this is the probability of *not* finding the problem set in drawer i given that it is in a different drawer j.

Using the Law of Total Probability, we determine

$$\mathbb{P}(\mathcal{B}_i) = \mathbb{P}(\mathcal{A}_i)\,\mathbb{P}(\mathcal{B}_i \,|\, \mathcal{A}_i) + \mathbb{P}(\mathcal{A}_i^{\mathrm{c}})\,\mathbb{P}(\mathcal{B}_i \,|\, \mathcal{A}_i^{\mathrm{c}}) = p_i q_i + 0 \cdot (1 - p_i) = p_i q_i.$$

Putting it all together, we get

$$\mathbb{P}(\mathcal{A}_j \,|\, \mathcal{B}_i^{\mathrm{c}}) = \frac{p_j}{1 - p_i q_i}.$$

Alternative method to compute the denominator $\mathbb{P}(\mathcal{B}_i^{\mathrm{c}})$: We can use the Law of Total Probability to compute $\mathbb{P}(\mathcal{B}_i^{\mathrm{c}})$ directly:

$$\begin{aligned}
\mathbb{P}(\mathcal{B}_i^{\mathrm{c}}) &= \sum_{k=1}^{n} \underbrace{\mathbb{P}(\mathcal{A}_k)}_{p_k} \underbrace{\mathbb{P}(\mathcal{B}_i^{\mathrm{c}} \,|\, \mathcal{A}_k)}_{\substack{1 \text{ if } k \neq i, \\ (1-q_i) \text{ if } k = i}} \\
&= \sum_{k \in \{1,\ldots,n\} \setminus \{i\}} p_k + p_i(1 - q_i) \\
&= 1 - p_i + p_i(1 - q_i) \\
&= 1 - p_i q_i.
\end{aligned}$$

(b) Determine the probability that the problem set is in drawer i.

Solution:

Based on the conditional probability law given $\mathcal{B}^{\mathrm{c}}$, we obtain

$$\begin{aligned}
\mathbb{P}(\mathcal{A}_i \,|\, \mathcal{B}_i^{\mathrm{c}}) &= 1 - \sum_{j \in \{1,\ldots,n\} \setminus \{i\}} \mathbb{P}(\mathcal{A}_j \,|\, \mathcal{B}_i^{\mathrm{c}}) \\
&= 1 - \sum_{j \in \{1,\ldots,n\} \setminus \{i\}} \frac{p_j}{1 - p_i q_i} \\
&= 1 - \frac{1 - p_i}{1 - p_i q_i} \\
&= \frac{p_i(1 - q_i)}{1 - p_i q_i}.
\end{aligned}$$

Alternatively, we can use Bayes' rule:

$$\mathbb{P}(\mathcal{A}_i \,|\, \mathcal{B}_i^{\mathrm{c}}) = \frac{\mathbb{P}(\mathcal{A}_i)\,\mathbb{P}(\mathcal{B}_i^{\mathrm{c}} \,|\, \mathcal{A}_i)}{\mathbb{P}(\mathcal{B}_i^{\mathrm{c}})} = \frac{p_i(1-q_i)}{1-p_i q_i},$$

where we used that $\mathbb{P}(\mathcal{B}_i^{\mathrm{c}}) = 1 - p_i q_i$ that we established in part (a).